PAPER_889: E(t) Vs ΛCDM Dark Energy Contrast

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-08 **Session:** 209 **Source:** Sessions 204-208 standalone module integration **Calculator:** EtVsLambdaCDMDarkEnergyContrastCalc (CP4 #473) **CVW:** v2.0.0 compliant

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Abstract

Direct comparison of UQFF E(t) equation of state against ΛCDM w=-1 cosmological constant. Analyzes fine-tuning: QFT/observed $\Delta\Lambda$ ~ 10^{120-139}. UQFF resolves this with 2 calibrated parameters (κ, [SSq]) vs ΛCDM's static cosmological constant. 5-row contrast table covering equation of state, dark energy density, fine-tuning, physical origin, and time evolution.

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1. Core Equations

```
w_ΛCDM = -1
w_UQFF from E(t)
ρ_Λ = 0.692·ρ_crit
fine-tuning ~ 10^120
```

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2. Parameters

Parameter	Default	Description
E_0	1.0 J	Initial energy scale
F_UBi_over_FU	0.8	Buoyancy-to-field ratio

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3. UQFF Integration

This calculator integrates et_full_lagrangian.py from Session 206 into the CP4 pipeline as class #473.

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